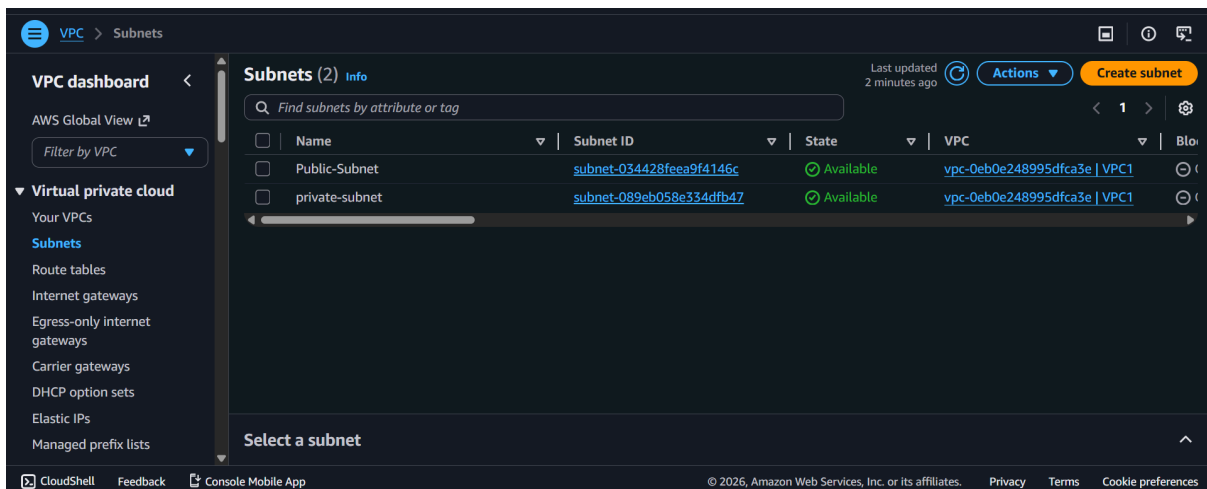
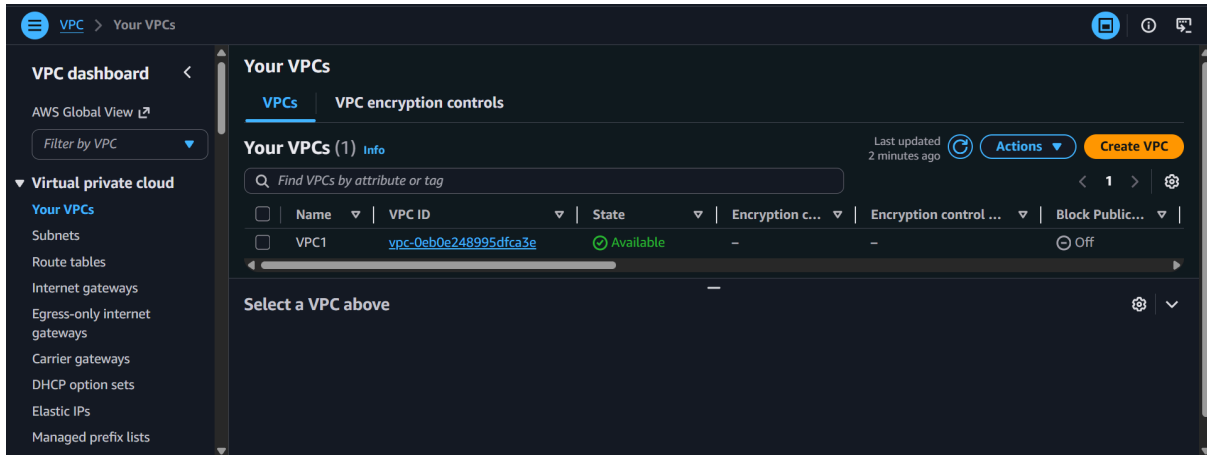


1. Creating the VPC and Subnets

- Navigates to the AWS Console and creates a custom VPC named vpc1 with the CIDR block 10.0.0.0/16.
- Inside this VPC, creates two separate subnets:
 - Public Subnet (CIDR: 10.0.0.0/24)
 - Private Subnet (CIDR: 10.0.1.0/24)



2. Configuring Internet gateway and Route Table

- **Public Route Table:** Configures a route for 0.0.0.0/0 pointing to the IGW and associates it with the Public Subnet.
- **Private Route Table:** Associates it with the Private Subnet (initially left without any internet route).

Internet gateways (1/1) Info

Name	Internet gateway ID	State	VPC ID	Owner
INTERNATE GATEWAY	igw-0183ab83c9fd0ab51	Attached	vpc-0eb0e248995dfca3e VP...	610356898178

igw-0183ab83c9fd0ab51 / INTERNATE GATEWAY

Route tables (4) Info

Name	Route table ID	Explicit subnet associ...	Edge ass...	Main	VPC
-	rtb-07c2c2893859b1e99	-	nat-14bd5f7...	No	vpc-0eb0e248995dfca3
Private Route Table	rtb-0bde12fde08b099d	subnet-089eb058e334df...	-	No	vpc-0eb0e248995dfca3
-	rtb-02cbaf4f188ab2fde	-	-	Yes	vpc-0eb0e248995dfca3
Public Route Table	rtb-03cbdc2c481bdc0f5	subnet-034428feca9f414...	-	No	vpc-0eb0e248995dfca3

Select a route table

rtb-03cbdc2c481bdc0f5 / Public Route Table

Details Info

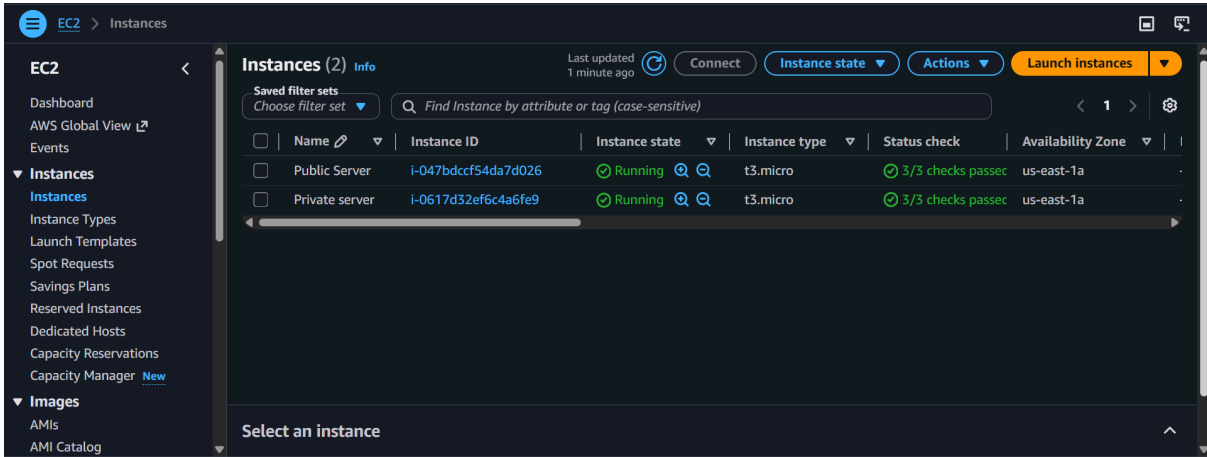
Route table ID rtb-03cbdc2c481bdc0f5	Main No	Explicit subnet associations subnet-034428feca9f4146c / Public-Subnet	Edge associations -
VPC vpc-0eb0e248995dfca3e VPC1	Owner ID 610356898178		

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0183ab83c9fd0ab51	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

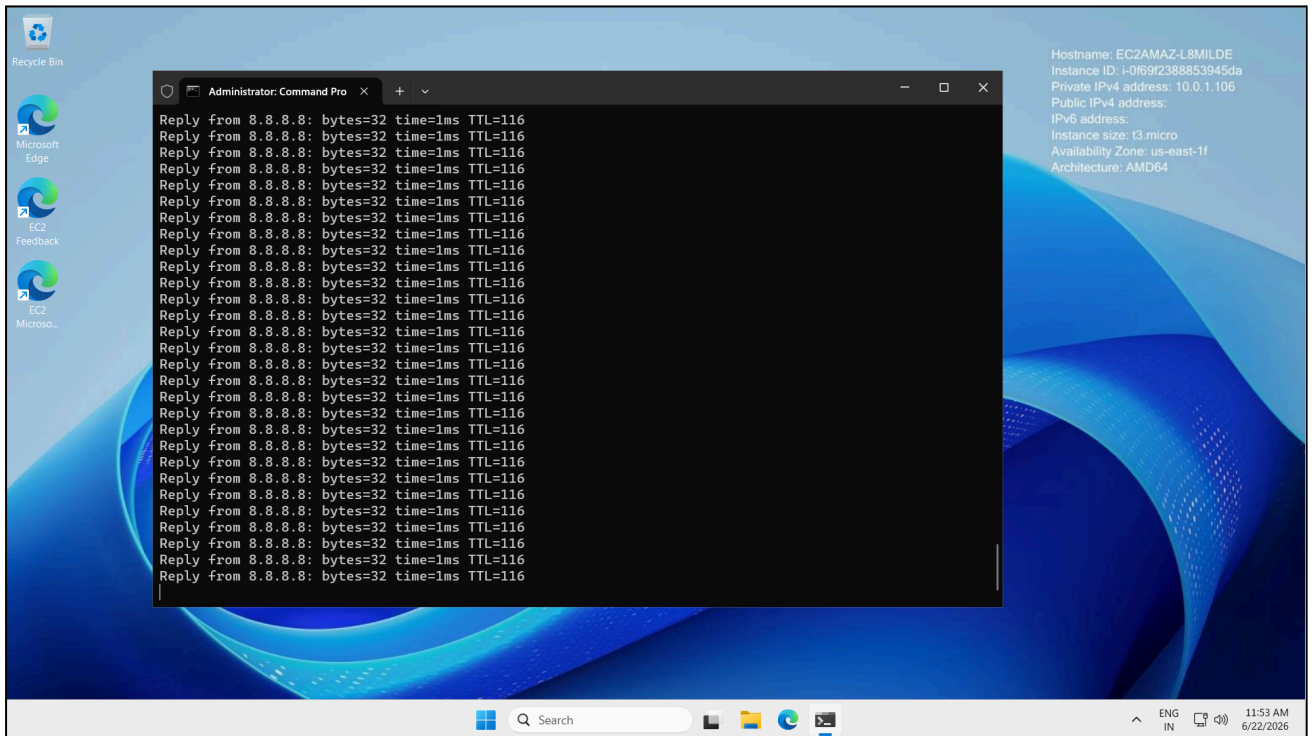
3. Launching EC2 Instances

- **Launches two Windows EC2 instances within the VPC:**
 - **Public Server:** Launched in the Public Subnet with Public IP Enabled.
 - **Private Server:** Launched in the Private Subnet with Public IP Disabled.



4. Connecting and Initial Internet Testing

- Connects to the Public Server first via Remote Desktop Protocol (RDP).
- From inside the Public Server (acting as a Bastion Host), opens another RDP session to connect to the Private Server using its internal Private IP.
- Opens the Command Prompt on the Private Server and runs `ping 8.8.8.8 -t`. It initially shows Request timed out because there is no internet access.



5. Creating the NAT Gateway & Updating Routes

- Navigates to the VPC Dashboard and clicks Create NAT Gateway.
- Crucial Step: Deploys the NAT Gateway inside the Public Subnet and allocates an Elastic IP (EIP) to it.
- Goes back to the Private Route Table, adds a route for 0.0.0.0/0, and points the target to the newly created NAT Gateway.

